

Parker Gray

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Summary

Computer Science Master's graduate looking for an exciting and challenging career in data engineering. Fields that interest me most are data science, artificial intelligence, embedded systems, and machine learning. I've explored a variety of subjects as a result of my education and have experience in web development, compiler design, firmware programming, game development, data analysis & engineering, amongst others. Languages I've used most commonly for development include C++, Java, and Python. I'm a self-starter able to develop solutions to problems on my own with little oversight.

Education

California State University East Bay, Hayward, CA	08/2019 – 12/2020
MS, Computer Science	
San Francisco State University, San Francisco, CA	08/2013 – 05/2018
BS, Computer Science, Minor in Mathematics	

Work Experience

Data Engineer, Cigna	08/2022 – 08/2023
• Worked amongst a team of engineers to deliver productionized Pyspark and SQL data pipelines that ingest incoming data, and output transformed data on weekly, monthly, and yearly cadences. • Solely responsible for the development and maintenance of several data models from start to finish, took over models from other employees, provided a supporting role to other models. • Familiar with the AWS environment's cloud services commonly used for data modeling including Glue, Redshift, S3, EMR, Cloudwatch Logs, and AWS Lambda Step Functions. Pilot tester of Cigna's Databricks cloud instance, converting AWS-ready models to Databricks, documenting necessary changes and runtime performance compared to AWS. I provided documentation to understand how to convert models from AWS to Databricks for my team members. • Two models I was solely responsible for: Presale – given patients from Cigna and their recent OTC drug and rx prescriptions, produce an output that scores which drugs and rx prescriptions as most commonly used while also predicting which prescriptions a patient might use next given this data. • Jumpstart – given an extract of medical claims from an external client who is interested in having their employees enter a Cigna medical plan, determine from these claims which patients present the most potential cost risk to Cigna by scoring them based on past and present high risk diagnoses.	
Data Engineer, Pacific Gas & Electric Company	06/2021 – 04/2022
• Constructed ETL models that ingest PG&E external tree contractor company datasets using Pyspark, SQL, Visual Basic for Applications, Apache Spark, and Pandas. Arranged output data in graphical format for presentation with PowerBI and PowerPoint. Drew conclusions from data and presented said conclusions to managers and executives. • Created a database to house historical weekly data from PG&E's tree contractors by leveraging an existing Microsoft SQL Server instance within the company. Collaborated with other teams to construct similar ETL models that efficiently ingest/output/present their own data. • Assisted data platform team with finding and squashing bugs related to the above server, follow up ETL models, and provided general feedback on improvement to PG&E's data processes.	

Projects

Internet of Things-Powered Homemade Security System – C++

- Created a microcontroller & sensor based homemade security system. Used Raspberry Pi and Arduino microcontrollers for the project, each with their own sensor attachment.
- One sensor determines if an object is present via heat sensor, the other sensor operates as a camera, taking a picture when heat sensor is tripped. The two microcontrollers send alerts to each other using an MQTT server hosted on RPi's Raspbian OS, sending signals and picture data to one another. Picture data is categorized into a database and displayed on a website. **I was the sole developer on this project, designing, coding, and debugging all of its attributes.**

Multiplayer Battle Royale FPS Game – C#

- Utilized the Unity development engine to create a fully featured battle royale FPS game, complete with multiplayer lobby system, typical FPS physics, and multiple game mode types.
- Project work consisted of utilizing C# to script game assets to represent the game world as well as designing and implementing a functional user interface for gameplay and network features. Utilized Agile framework and common DevOps practices for development of this project. **Project was developed by a team, of which I was one of three members. I performing coding and debugging for game physics and game networking related portions of the project.**

3-SAT Genetic Algorithm – C++

- Created a genetic algorithm to determine satisfiability of large datasets of the NPC 3SAT problem.
- The algorithm mimics evolutionary development, starting with a collection of possible answers ('genes'), judging them by their fitness as an answer, and moving to the next population by having highest fitness answers 'mate', randomly exchanging their data to create a child. The algorithm continues until a certain percentage of the answers in the population are deemed sufficiently fit as an answer to the question. **I was the sole developer on this project, designing, coding, and debugging all of its attributes.**

Connect 4 in LISP - Lisp

- Remade the famous Connect4 game in the functional programming language LISP. Players fight against an AI who thinks many moves ahead.
- Large portion of the project is dedicated towards implementing a proper heuristic for the AI to follow for the Connect 4 game. The AI's decision-making process ultimately resembles A* search, picking the highest utility move given a gameboard state based on how valuable that move is when considering future board states. **I was the sole developer on this project, designing, coding, and debugging all of its attributes.**

Technical Skills

Tools: Apache HTTP Server, Apache Spark, Amazon Web Services, CLI, Docker, Eclipse, Git, Linux, MATLAB, Microsoft Azure, Microsoft Excel, Mongo, MySQL, NPM, Pandas, PostgreSQL, PowerBI, PowerPoint, PyTorch, TensorFlow, Unity, VSCode

Languages: C, C++, C#, CSS, HTML, Java, Javascript, LISP, PHP, Python, R, Ruby

Skills: Multi-threading, Data analysis, Data pipelining, Extract-Transform-Load operations, Code debugging, Embedded systems, microcontrollers